SA 220 AQUACULTURE

❖ FEW ANNOUNCEMENTS

- Classes will be commencing on the stipulated time and for the stated duration
- You are encouraged to attend classes

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❖ **Course Materials:** You will be provided with the following materials.

➢ Lecture notes after the material has been delivered.

➢ List of recommended textbooks and relevant articles.

❑ Not compulsory for you to acquire or indeed read, But will help to supplement the course material.

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❖ **Assessment**

- There shall be a test in week two of learning (Agree on the date and time)
- We shall have an assignment (Group or individual)

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- **GETTING TO KNOW EACH OTHER**
- Self introductions (Everyone)
- What do you expect from this class?

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- Fisheries Officer (MFL)- Ndola
- MSc Sustainable Aquaculture (**Merit**) – University of Stirling, UK.
- MSc **candidate** Aquaculture and Fisheries Science – LUANAR, Malawi.
- BSc Fisheries and Aquaculture (**Distinction**) –CBU, Zambia.

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Course aim:

To provide the basic background information on Aquaculture with special reference to fish farming and also to link capture fisheries and aquaculture as contributing sectors to food fish as a cheap source of protein in human diets.

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Course objectives:

- To know the different key terms used in aquaculture
- To appreciate the importance of aquaculture to society and SDGs
- To know and identify the various fish culture system
- To know the species of economic importance and factors considered when choosing them.

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Course objectives:



- To have a basic knowledge of hatchery and pond management
- To explain the basics of fish feed formulation
- To know the food and feeding habits of different fish
- To understand the basics of artificial propagation of fish.

KEY TERMS

1. AQUACULTURE

- Aquaculture is the farming of aquatic organisms, including fish, mollusks, crustaceans and aquatic plants in a natural, semi-controlled or fully controlled manner

FAO (1990) https://www.fao.org/3/t0697e/t0697e02.htm

KEY TERMS

2. CRUSTACEANS



- Crustaceans are a very diverse group of invertebrate animals which includes active animals such as the crabs, lobsters, shrimp, krill, copepods, amphipods, and more sessile creatures like barnacles.

KEY TERMS

3. MOLLUSCS

Molluscs are soft bodied animals, often with hard shells for protection





KEY TERMS

4. FISH SEED OR FINGERLING

- A fish ‘seed’ is a juvenile fish to be used for fish farming



5. FISH BREEDING

- entails all the various techniques or methods used in fish seed production

KEY TERMS

6. BROODSTOCK/BROODER

- Brood stock, or brood fish are a group of mature individuals used in aquaculture for breeding purposes or brood fish, are the parent fish from which fry and fingerlings are produced

KEY TERMS

7. FISH HATCHERY

- A fish hatchery is a place where large numbers of fish eggs are artificially fertilized and fry are hatched in an enclosed environment.

KEY TERMS

8. HYPOPHYSATION

- Hypophysation is artificial or induced breeding, a technique that uses pituitary gland extractinjection



Plate 4. Ovatide: A synthetic hormone that can also be used.

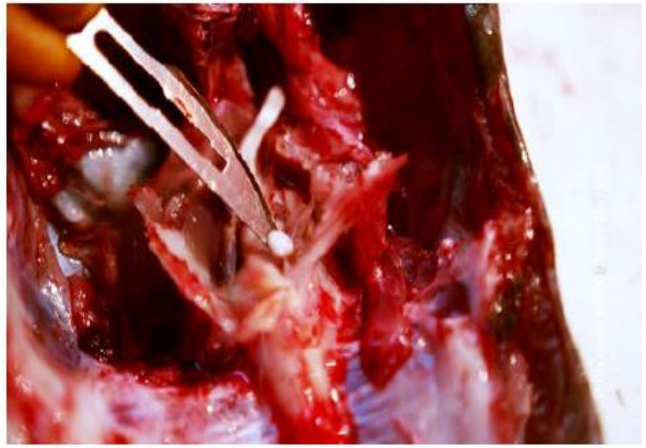


Plate 3. Pituitary gland being extracted.



Plate 5. Injecting a female broodstock above lateral line.

KEY TERMS

9. INGREDIENT

- A component part or constituent of any mixture making up a feed.

10. NUTRIENT

- The components in a diet that an organism utilizes to survive and grow.

KEY TERMS

11. FEED CONVERSION RATIO (FCR)

- This is the measures the efficiency with which an animal converts feed into its body mass.

Formulae:

FCR=Amount of Feed given / Amount of weigh gained by the animal

KEY TERMS

12. Natural fish food organisms - plankton, insects and other aquatic organisms that fish eat.

13. Phytoplankton - the plant component of plankton.

14. Plankton - the various, mostly microscopic, aquatic organisms (plants and animals) that serve as food for larger aquatic animals.

KEY TERMS

15. Pond dike - the wall of a pond which is constructed to hold in the water.

16. Predatory/carnivorous fish - a fish species that eats other fish.

17. Supplemental/incomplete feed - a feed that does not contain all the vitamins and nutrients essential for growth, and which is produced outside of the pond.

KEY TERMS

18. Omnivores- These are species that feed mainly on both plant and animal materials. Most Catfish species are mainly omnivores with diets ranging from algae, vegetable matter and insects.

19. Herbivores- These are the fishes that feed predominantly on higher plant materials including vegetable matter.

20. Monoculture - This is the culture of only one species of fish.

KEY TERMS

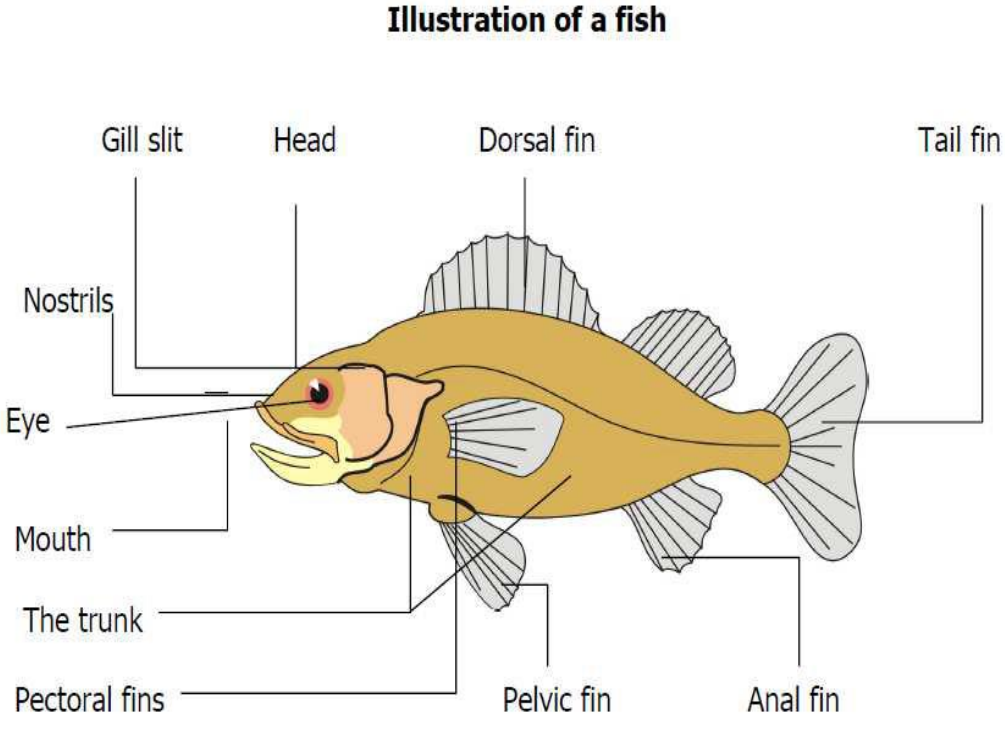
21. Monosex culture - This is the farming of only one sex (Male or Female) done by fish culturists through hormonal use, hand sexing or the use of supermales (YY) or superfemales (XX).

22. Polyculture or composite fish culture- It is a type of fish culture in which different types of fishes of different feeding habits are Cultured.

Points of Discussion

1. **What is a Fish ?**







Points of Discussion

1. **What is the difference between Fish and Fishes?**



Points of Discussion

1. What are the main sources of broodstock for hatcheries ?
2. Is it possible to successfully do aquaculture without utilizing wild resources?
3. Which one is better. A high FCR or Lower FCR?

Answers

1. **Wild (natural water bodies) and Hatchery reared**

2. **No. We need the wild for genetic diversity**

3. **A lower FCR is better**

(A lower FCR indicates better efficiency (utilisation) because it means the animal requires less feed to gain a certain amount of weight. High FCR values can signify inefficiencies in feed utilization, which can increase production costs and environmental impacts in aquaculture operations. Thus, aquaculture farmers aim to optimize FCR to improve profitability and sustainability)

Answers

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TAKE HOME QUESTION

Differentiate the following terms

- Aquaculture and fish farming
- Aquaculture and Agriculture

END

THANK YOU FOR YOUR ATTENTION